THALES

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Speciality Engineering Plan

562

RAM Plan

Generic

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1 SCOPE

1.1 Purpose

This document defines RAM activities for the whole product life cycle in a generic way. The contents are applicable to all railway-based systems such as Interlocking, trackside and on-board systems of the GTS product portfolio which have to cope with RAM issues according to the CENELEC standard EN 50126-1 [S1] and related standards.

The assignment between requirements of EN 50126-1 [S1] and RAM activities defined in this generic RAM plan is shown in the attached excel document [11].

This document provides the audience with the information on the major processes necessary to guarantee an acceptable level of dependability throughout the whole product's lifetime. It supports RAM activities as a guideline on methods and tools for any project RAM-AM.

1.2 Intended Audience

The intended audience of this generic RAM Plan are the following functions within a project:

- Product Line Manager
- Project Design Authority
- Safety Assurance Manager
- Solution Engineering Manager
- RAM-Assurance Manager
- Quality Assurance Manager
- Project Manager
- Documentation Manager
- Industrial Engineer
- Service Engineer

Responsibilities and description of the roles and activities can be found in the Project Life Cycle (see [S12]) together in accordance to the roles as defined in EN 50126-1 [S1], EN 50128 [S2] and EN 50129 [S3].

1.3 Relationship to other Plans

This generic RAM Plan should be addressed within the PMP. It presupposes a proper quality assurance described in the QMP. The generic RAM Plan is a configuration item controlled and managed following the specification of the CMP.

- The Project Management Plan (PMP) describes how the project is executed, monitored and controlled, and finally closed.
- The Quality Management Plan (QMP) describes the application of quality assurance management. It is a basis for the verification activities within the project.
- The Configuration Management Plan (CMP) describes configuration management processes and procedures, including baseline management.
- The Safety Plan describes the application of safety assurance management and how safety in sense of applied CENELEC standards has to be implemented.

1.4 Document Management

The management of this document will be performed according the Document management Rules [S13].

1.4.1 Change History

1.4.1.1 Edition 03

This generic RAM Plan was based on the Development Life Cycle DLC. The current edition has been updated to follow the new process model Chorus Thales Group Reference System and Governance Tool.

1.4.1.2 Edition 04

- References updated.
- Updated according to the latest version EN 50126-1:2017 [S1]. For a detailed overview of the changes of EN 50126-1:2017 with respect to its predecessor version, please refer to [10].

1.4.1.3 Edition 05

- References updated.
- Template updated.
- Definition of MRT added in 2.3 and 5.3.2.6.
- Release by signature.

2 References and Terms

2.1 Referenced Documents

Id	Reference
[1]	Reliability Standard Reliability Classes 1AA 00210 0002 DSZZA, Ed. 03
[2]	Reliability Standard Reference Failure Rate 1AA 00210 0003 BCZZA, Ed. 03
[3]	Reliability calculation methods for telecommunication systems 1AA 00210 0005 DSZZA, Ed. 01
[4]	Dependability Performance Objectives 1AA 00210 0006 ASZZA, Ed. 02
[5]	User manual for the AlcaRel reliability program 1AA 10001 AAAA PCZZA, Ed. 08
[6]	Reliability Standard Failure Rate Factor 1AA 00210 0001 BCZZA, Ed. 04
[7]	Spare parts calculation 1AA 00210 0004 ASZZA, Ed. 02
[8]	AlMaTo - THALES Markov Analysis Tool User Manual 1AA 10072 AAAA PCZZA, Ed. 01
[9]	Assure RAM/LCC Process 83530293-DDQ-AUT-EN-003
[10]	Analysis Report, Changes in CENELEC standard, EN 50126:2017, EN 50129:2018, EN 50128:2011; 3BU 15000 0612 BEAPA, Ed.03
[11]	RAM Plan Generic Excel Attachment 3BU 15000 1712 DUAPA, Ed.05

Table 1 : Referenced Documents

2.2 Standards and Procedures

Id	Reference
[S1]	EN 50126-1:2017 - Part 1: Railway application – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS), Part 1: Generic RAMS Process
[S2]	EN 50128:2011 + A1:2020 + A2:2020 - Railway application – Communication, signalling and processing systems – Software for railway control and protection systems
[S3]	EN 50129:2018 + AC:2019 - Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling
[S4]	IEC 60300:2014 - Dependability Management
[S5]	IEC 60706:2007 - Guide on Maintainability
[S6]	IEC 60812:2018 - Analysis techniques for system reliability- Procedures for failure modes and effect analysis (FMEA)
[S7]	SN 29500: Siemens Standard
[S8]	IEC 61025:2006 - Fault tree analysis (FTA)
[S9]	IEC 61165:2006 - Application of Markov techniques
[S10]	IEC 60605:2007 - Equipment reliability testing
[S11]	Reparaturenlenkung 3BU 00600 0010 PDAPC, Ed. 03
[S12]	Project Life Cycle 83530275-DDQ-AUT-EN-004

Id	Reference
[S13]	Document Management Rules, 83530061-DOC-AUT-EN, Rev. 011
[S14]	FIDES guide 2009, Edition A, September 2010, Reliability Methodology for Electronic Systems
[S15]	MIL-HDBK-217F - Reliability Prediction of Electronic Equipment 1AA 00230 0001 ASZZA, Ed. 01

Table 2 : Standards and Procedures

2.3 Terms, Abbreviations and Acronyms

CENELEC	European Committee for Electrotechnical Standardisation
CCB	Configuration Control Board
CMP	Configuration Management Plan
COTS	Commercial Off-The-Shelf
DPR	Dependability Requirements
EMC	Electromagnetic Compatibility
ESS	Environmental Stress Screening
FMEA	Failure Modes and Effects Analysis
FRACAS	Failure Reporting And Corrective Action System
FTA	Fault Tree Analysis
GTS	Ground Transportation Systems
HAZOP	Hazard Operational Analysis
IEC	International Electrotechnical Commission
LCC	Life Cycle Cost
LRU	Least repairable/replaceable unit
MRT	Mean Repair Time
MTBF	Mean Time Between Failures
MTTR	Mean Time To Restore
PBA	Printed Board Assembly
PMD	Procure, Make & Deliver
PMP	Project Management Plan
QMP	Quality Assurance Plan
RAM	Reliability, Availability and Maintainability
RAM-AM	RAM Assurance Manager
SAM	Safety Assurance Manager
SSS	System / Segment Specification
TAT	Turn Around Time

Table 3 : Terms, Abbreviation and Acronyms

2.4 Definitions

Availability:

The ability of a product to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval assuming that the required external resources are provided.

Common Cause Failures:

Failures of multiple items, which would otherwise be considered independent of one another, resulting from a single cause.

Error:
A discrepancy between a computed, observed or measured value or condition and the true, specified or theoretically correct value or condition.

Failure:
The termination of the ability of an item to perform a required function. After failure the item has a fault. Failure is an event, as distinguished from fault, which is a state.
Note: Failures are categorized as systematic or random

Systematic Failure:
A failure due to errors, which cause the product, system or process to fail deterministically under a particular combination of inputs or under particular environmental or application conditions.

Fault:
The state of an item characterized by the inability to perform a required function, excluding the inability during preventive maintenance or other planned actions.

Intermittent Fault:
An intermittent fault is one, which sporadically exhibits erroneous behaviour and then correct behaviour.
Permanent Fault:
A permanent fault is one, which is stable and continuous. Permanent hardware failures require some component to be replaced or repaired.
Transient Fault:
A transient fault cause erroneous behaviour for a short period of time and then disappears.

Fault detection:
All mechanisms aimed at recognizing that something unexpected has occurred. Detection strategies are broken into two major categories, online detection and offline detection.
Fault diagnosis:
All mechanisms aimed at determining the cause(s) of error(s), in terms of both location and nature.
Fault passivation:
All mechanisms aimed at preventing the component(s) identified as being faulty from being invoked in further executions.

Fault tree:
A logic diagram showing which fault modes of sub-items or external events, or combinations thereof, result in a given fault mode of the item.

Maintainability:
The ability of an item under given conditions of use, to be retained in, or restored to, a state in which it can perform a required function, when maintenance is performed under given conditions and using stated procedures and resources.

Maintenance:
The combination of all technical and administrative actions, including supervision actions, intended to retain an item in, or restore it to, a state in which it can perform its required function.

MRT:
The MRT (**M**ean **R**epair **T**ime) represents the time required for repair of the item that is out of service, without considering administrative or logistics delay.

MTBF:
The MTBF (**M**ean **T**ime **B**etween **F**ailures) is the expectation of the operating time between failures. MTBF is usually reserved for items with *constant* failure rate.

MTTR:
The MTTR (**M**ean **T**ime **T**o **R**estore) expectation of the time to restoration. MTTR is the expected value (mean) of an item's repair time.

Turn Around Time:
The Turn Around Time (TAT) is the time span starting with the failure of a unit and the replacement by a spare part until the repaired part is available on stock again.

Railway duty holder:

Body with overall accountability for operating a railway system within the legal framework.

Reliability:

The ability of an item to perform a required function under given conditions for a given time interval.

3 RAM Management

3.1 The acronym RAM

RAM stands for

- **R**eliability
- **A**vailability
- **M**aintainability

Reliability states the ability of an item to perform a required function under given conditions for a given period of time. It only considers repairable units when not leading to any service interruption of the item to be considered, whereas availability also covers down times of the item itself. Maintainability is the ability that a maintenance action can be completed within a specified time span. The maintenance actions can be either preventive or corrective, respectively. All three quantities are strongly interdependent and so they are treated together.

RAM in conjunction with logistic support represents the dependability of a product. So, dependability can be seen as the ability of a product to deliver its specified level of service to its users especially in the presence of failures, which have an impact on the level of service. In this context the term dependability does not cover further attributes of railway signalling systems such as safety, integrity etc. Please refer to [4], [S4] and [S5] for details.

3.2 Main Tasks of the RAM Management

The main tasks of the RAM Management are

- Specification of RAM requirements
- Determination of standards and procedures
- Recording and prevention of negative impacts on RAM characteristics
- Planning and implementation of RAM activities
- Demonstration of the achievement of RAM requirements
- Maintenance of the tool chain for the demonstration of the achievement of RAM requirements
- Permanent monitoring of RAM-specific performance throughout the product life cycle.

3.3 Additional tasks

3.3.1 Environmental Tests of system components

The adequate suitability for use of system components is a precondition for reaching a sufficiently high level of reliability. Therefore, it must be of considerable interest to any RAM-AM. So, the RAM Management has to check the adequacy and completeness of the measures to be taken against environmental influences and the RAM-AM has to play an active role concerning the definition of measures against environmental influences and the monitoring of the realisation thereof. See also chapter [5.2.4]. This holds for tests within the development phase as well as for environmental tests to check the degree of fitness before starting industrial production. At this stage environmental tests are a kind of first article inspection (FAI) and therefore the RAM-AM has also an interface to PMD.

3.3.2 RAM-based LCC

Life cycle cost is an ever-increasing important factor not only during the internal decision-making process, but also as a substantial criterion within a tendering process. Certain dependability predictions have an impact on this cost package. This contribution is described in chapter [5.7].

3.4 RAM Assurance Manager

Right from the start it must be assured that all RAM aspects are taken into account. Therefore, a nominated RAM Assurance Manager[1] (RAM-AM) must accompany each project and participate in all essential meetings and reviews. The RAM-AM should be a representative from the RAM Management. If there is no RAM Management in the organisational structure the RAM tasks have to be taken over by a person from another organisational unit, preferably by the quality management. Nevertheless, a RAM-AM has to be foreseen. Furthermore, the RAM-AM must be a permanent member of the CCB (Configuration Control Board) established to control all changes to the requirements.

The RAM-AM ensures

- that RAM procedures are defined and followed correctly
- that the product meets the related RAM requirements
- the specification of RAM requirements
- the planning and implementation of RAM activities
- demonstration of the achievement of RAM requirements
- a permanent monitoring of RAM-specific performance

[1] In the scope of the GTS management system framework the RAM/Safety management is a subset of the SAM responsibilities.

4 RAM activities during the life cycle of a product

4.1 Life cycle of a product

The life cycle of a product or system is the sequence of phases and the corresponding activities during the whole lifetime, reaching from the first conceptual considerations up to the decommissioning and disposal of the product.

The RAMS process as described in [S1] consists of three major blocks that represent the life cycle sequentially and are additionally connected among themselves with a feedback loop and subsequent loops:

- Risk assessment:
- Implementation and demonstration of compliance with requirements
- Operation, maintenance and decommissioning

The list below shows the individual phases according to CENELEC 50126-1 [S1]. A detailed allocation of all RAM activities to each phase can be found in the attached excel document (see [11]).

1. Concept
2. System definition and operational context
3. Risk analysis and evaluation
4. Specification of system requirements
5. Architecture and apportionment of system requirements
6. Design and implementation
7. Manufacture
8. Integration
9. System validation
10. System acceptance
11. Operation, maintenance and performance monitoring
12. Decommissioning

4.2 Phase related RAM activities

Depending on the organisational/ management structure of the system under consideration, all relevant stakeholders such as railway duty holder and railway supplier have to be identified and their responsibilities within the life cycle phases have to be defined.

The first block “Risk assessment” which includes phases one to four, is part of the responsibilities of the railway supplier, whereby the phases within the block “Implementation and demonstration of compliance with requirements” are the central RAM phases and in the responsibility of the railway duty holder. For an overview of all phases refer to Figure 6 in EN 50126-1 [S1].
Since all phases are additionally connected to each other by feedback and subsequent loop, it is important to ensure that responsibilities are precisely assigned.

4.2.1 General

In the following subchapter a listing of RAM activities is for each single phase of the product’s life cycle elaborated according to the CENELEC standard EN 50126-1 [S1]. The requirements in this standard are stated in a generic way in order to be applicable to a variety of railway-based applications. Therefore, the application of these rules has to be adapted adequately to the particular requirements of the system under consideration. The determination of the scale and the depth of the application are supported by the creation of action packages to each phase related activity. In any case, the adequacy of the actions chosen have to be justified and the decisions made have to be recorded in written form.

All RAM activities should adhere to the rules listed below:

- When dealing with dependability issues the maxim should be “as good as necessary” rather than “as good as possible”.
- The activities necessary to ensure the required degree of dependability should be carried out continuously during the life cycle phases.
- The activities must be performed in co-operation with all project roles.
- Co-ordination and controlling of RAM activities are directed by the central function RAM Management.

4.2.2 Listing of the phase-specific RAM activities

- Concept
  - Review previously achieved RAM performance
  - Consider RAM implications of the project Review
- System definition and operational context
  - Evaluate past experience data for RAM
  - Establish boundaries and functions of the system under consideration
  - Set RAM policy
  - Identify long-term operation and maintenance conditions
  - Identify influence on RAM of existing infrastructure constraints
- Risk analysis and evaluation
  - Identify and classify RAM equivalents of hazards associated with the system
- Specification of system requirements
  - Specify system RAM requirements
  - Define RAM acceptance criteria
  - Define system functional structure
  - Establish activities to fulfil RAM related requirements
  - Retrieve information to fulfil railway duty holder related requirements
- Architecture and apportionment of system requirements
  - Apportion system RAM requirements
    - Specify subsystem and component RAM requirements
    - Define subsystem and component RAM acceptance criteria
  - Establish project specific RAM Plan
- Design and implementation
  - Implement RAM program by review, analysis, testing and data assessment, covering
    - Reliability and availability
    - Optimal maintenance policy
    - Logistic support
  - Undertake Program control, covering
    - RAM program management
    - Control of subcontractor and supplier
- Manufacture
  - Perform environmental stress screening as applicable
  - Perform RAM-centered assurance arrangements
  - Commence failure reporting analysis and corrective action system FRACAS
- Integration
  - Start maintainer training
  - Initiate system support arrangements related to RAM (spare parts, tool provision)
- System validation

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  - Perform RAM demonstration
  - Ensure that RAM related requirements are covered in validation report
- System acceptance
  - Assess RAM demonstration
- Operation, maintenance and performance monitoring On-going procurement of spare parts and tools
  - Perform on-going reliability-centred maintenance logistic support
  - Collect, analyse, evaluate and use performance and RAM statistics
  - Maintain FRACAS process as applicable
- Decommissioning
  - Identify RAM impact of decommissioning.

4.2.3 Deliverables

In each project the following documents should be written, reviewed and archived:

- RAM concept (recommended)
- HW-FMEA (with focus RAM; recommended)
- HAZOP (with focus on operational RAM aspects; recommended)
- Reliability prediction report (recommended)
- Preliminary RAM analysis (recommended)
- RAM analysis (mandatory)
- FRACAS (if appropriate / required)
- RAM demonstration report (if appropriate / required)

Above all, a RAM concept is necessary when dealing with systems where fundamental ideas and approaches have to be communicated to the railway duty holder. Furthermore, it might be an integral part of the tender to emphasise the RAM aspects. As a minimum, the document should include a description of the architecture of the system including all redundancy mechanisms employed (if any) and if possible, the conceptual dependability aims in terms of quantitative predictions.

The HW-FMEA serves as a basis for the calculation of dependability quantities (downtimes, MTTR, etc.) and typically accompanies a RAM analysis. It also analyses diagnostic capabilities of the system in order to support preventive and corrective maintenance actions. Depending on the size and complexity of the analysis it might be convenient that the results of the FMEA can be presented in a separate document. For a detailed description please refer to [S6].

The HAZOP addresses those failure modes from HW-FMEA which have a significant impact on the operational behaviour in terms of loss of service or material.

The reliability prediction report presents reliability figures of electronic components on PBA level, based on the calculation methods described in section 5.3.2.4.

A preliminary RAM Analysis can be performed to support targets in complex projects. It should give a first impression of the expected failure rates and other RAM key figures of the system under consideration.

Any RAM analysis should at least include the following predictions:

- Failure rates of all PBAs and devices used
- MTBF and availability of all subsystems
- MTBF and availability of the overall system (maximum field relevant configuration).

The results have to be checked against the quantitative dependability requirements stated in the requirement specifications (SSS), respectively.

Any FRACAS as a minimum shall be populated with information about failures and defects identified during operation and maintenance.

Before releasing above documents a proposal must be distributed for review at least to the following project members

- Product Line Manager
- Project Design Authority
- RAM-Assurance Manager
- Safety Assurance Manager
- Solution Engineering Manager
- Quality Assurance Manager
- Project Manager
- Requirements Manager
- Documentation Manager
- Verifier
- Validator

Note: The excel file attached to this document [11] provides an overview of RAM documents, tasks and deliverables to be provided resp. recommended in the phases within the block “Implementation and demonstration of compliance with requirements”.

5 Generic guidelines

5.1 General

During the system requirements phase dependability requirements (DPRs) have to be defined. All DPRs are then listed in the SSS (System / Segment Specification). This document has to be seen as a top-level document and serves as a basis for further apportionment.

In the following chapter RAM relevant key issues are stated.

5.2 Dependability requirements

5.2.1 Principle of balance

When elaborating hardware dependability requirements, it is essential to keep in mind that the object to be analysed is typically a part of an entire system and cannot be considered being isolated from its environment. Consequently, a balance between the failure behaviour of the item and the impact of various other error sources has to be reached. Service interruptions can generally be caused by failures of peripheral elements, environmental influences and also by human errors. Failures of technical railway facilities are only a part of this cluster of possible failure sources and can be attributed to two main reasons: hardware faults and software errors.

From the reliability engineering point of view the major objective is to minimize failures. But, it is also known from experience that software errors are not completely avoidable. Although, they are typically rare events, the frequency increases with the degree of complexity of the system. This allows an estimation of the frequency of such undesired events, which are independent from the architecture to be chosen. Besides other aspects, these frequencies should serve as a basis for the choice of the system hardware dependability requirements. The maxim should always be “make a system as good as necessary” rather than “as good as possible” [S1], [S2], [S3]).

5.2.2 Quantitative requirements

Dependability figures will either be taken from the railway duty holders requirements or have to be determined by the responsible RAM Manager.

In the latter case the principle of balance has to be taken into account. In addition to that, one should keep in mind that the determining factor in railway applications is availability (readiness for service) rather than reliability (continuity of service), because it also takes the resulting operational consequences into account. That means that the sole specification of frequency-dependent dependability quantities such as MTBF and reliability without specifying the impact on service should be avoided. A consequence class should always accompany it.

A possible classification of severity levels could be

- Significant
- Major
- Minor.

The exact description of each class has to be defined in accordance with the responsible railway authority.

The quantitative requirements should at least include

- Overall system MTBF incl. grouping in consequence classes
- Overall system availability

5.2.3 Qualitative requirements

In order to guarantee the required level of service a qualitative requirement package has to be established.

Special attention has to be paid to the following RAM issues

- Independence of external systems in the presence of fault
- Modularisation (all levels, degree of modularisation incl. internal independence in the presence of faults, definition of the degree of the least repairable/replaceable units etc.)
- Static or masking Redundancy (n-modular redundancy, Error correcting codes etc.)
- Dynamic redundancy (standby redundancy incl. switch-over handling, recovery mechanisms etc.)
- Wear out failures (prevention, preventive maintenance etc.)

Note that the term fault stands for the entire fault class including permanent and temporary (transient and intermittent) faults.

Moreover, measures to ensure problem-free removal in the presence of faults have to be requested. The main points of the fault removal process to be treated are:

- Fault prevention
- Fault detection
- Fault diagnosis
- Fault passivation
- Fault correction

For a definition of terms regarding faults refer to chapter 2.3.

5.2.4 Environmental requirements

Besides the dependability requirements it has to be checked if the environmental requirements are stated adequately. The main issues here are

- Electromagnetic compatibility
- Climatic environmental conditions
- Mechanical environmental conditions.

5.3 RAM analysis

5.3.1 General

The RAM analysis serves as a theoretical means for demonstrating the fulfilment of the RAM specific requirements concerning system hardware. It will be realised as:

- Reliability Analysis
- Availability Analysis
- Maintainability Analysis

The calculations of reliability quantities such as MTBF (Mean Time Between Failures) and unavailability of systems are based on the failure rates of printed board assemblies and devices. Besides the usage of this complete tool-chain, which will be presented in the next chapter, we also accept reliability data from manufacturers, provided that the basis of calculation is clearly stated. (See also [S10]).

5.3.2 Tools and procedures

5.3.2.1 General

When performing reliability analysis, a variety of well-proven procedures and tools is applied. In the following chapters the essential ones are presented, which allows us performing reliability calculations of entire systems starting from the failure rates of the underlying electronic and electro-mechanical components. A comprehensive description of this and other procedures can be found in the documents listed in 2.2.

5.3.2.2 Fault class

No systematic faults such as design faults, production faults and also faults caused by environmental influences and mishandling are considered. Only randomly occurring hardware failures are taken into account. Faults of a systematic nature are minimized by following the rules of CENELEC during the development process and are not treated quantitatively (in accordance with the corresponding CENELEC standard).

5.3.2.3 Relationship to safety analysis

When performing safety analysis such as hazard analysis the same tool chain is used. As a consequence, it is important to note that the results of a reliability analysis, especially at the component level, serves as a basis for further safety considerations. This means that conservative approaches should be used at all steps of the analysis.

5.3.2.4 Calculation of components failure rates

The failure rates of components have to be calculated by following company's standard method [1], [2], [3] described in the document "Reliability Standard Failure Rate Factor" [6] and related documents and standards [S15], [S14], [S7].

This method is principally based on the models of the military handbook MIL-HDBK-217F[S15]. In order to avoid too conservative data, a corrective factor (Alcarel-factor) is applied reflecting experiences of the failure behaviour in field.

This handbook was published by the Department of Defence, Washington and is a collection of failure rates of all types of electronic and electro-mechanical components. The failure rate of each component is determined depending on:

- Component type
- Component complexity
- Environmental conditions
- Electrical stress
- Environmental stress screening
- Manufacturing quality
- Maturity of the manufacturing process

A similar, more modern approach and proper for comparison is the Fides 2009 standard [S14].

5.3.2.5 Dependability Calculations of PBAs and devices

The calculation of the overall failure rate of PBAs and devices must be done by means of the so-called part stress method. This military handbook term means that the failure rate is determined by summing up all components, whereby the whole set of parameters of the MIL-HDBK, as previously presented, is used to evaluate the failure rates.

The analysis program ALCAREL was developed especially for company's wide usage to support this procedure. Please refer to [5] for details.

5.3.2.6 Mean time to restore / Mean repair time

The MTTR represents the expected duration the item is out of service because of a failure. It is measured starting from the time the failure occurs until the time the item is restored to full operation. Since it strongly depends on railway duty holder activities such as alarming and travelling time of maintenance personnel it has to be defined in accordance with the corresponding railway authority. It should be noted that for many safety considerations this is the time period in which a second fault could lead to a safety critical system state, and therefore it should be kept reasonably short.

The MRT represents the time required for repair of the item that is out of service, without taking into account administrative or logistics delay.

Please refer to figure B.1 of [S1] for details.

5.3.2.7 Dependability calculations at the system level

At system level quantitative analysis should be carried out with the help of a fault tree analysis tool to determine dependability quantities such as MTBF, reliability and availability (see [S8], [8]). This method allows the analysis of systems with a large number of components.

Besides the calculation of the above-mentioned quantities a fault tree analysis tool provides you with additional features such as importance and sensitivity analysis, which enables one to perform a detailed system examination.

For complex problems (in a mathematical sense) the Markov tool AlMaTo or an equally appropriate tool should be used. For more information refer to [8] and [S9].

5.4 Early stage failures

5.4.1 General

Typically, quality aims are reached if an item passes the acceptance sampling test. That means that freedom from fault - up to a certain extent – is guaranteed when putting a system into operation. But this doesn't automatically mean that all items will show a similar failure behaviour in the field. It is therefore the task of the RAM Management to ensure that during the initial operational phase no comparatively high number of failures - so-called early stage failures- will occur. Early stage failures have to be distinguished from failures that occur when putting a system into operation, and therefore they need a special treatment.

Generally, the failure rate of a population of statistically independent, identically distributed items shows the typical form of a bathtub, which can be divided into 3 sections. The first section shows the failure rate decreasing with time. As a rule, failures at this time can be attributed to randomly distributed weaknesses concerning materials, components or manufacturing processes etc.

In the worst case the early stage failures last up to a few thousand operation hours, depending on the complexity of the item to be considered and the maturity of the manufacturing process. In order to prevent early stage failures, there is the possibility of a longer run-in time under normal environmental conditions. Another approach would be a pre-treatment under extreme environmental conditions, a so-called "Environmental Stress Screening" (ESS) at PBA level. During a pre-treatment the test object is subjected to shocks and strains, such as thermal cycles and vibrations, to provoke early stage failures. Basically, this is a complex procedure because various technologies are tested simultaneously and they do react differently to those strains. So, unwise pre-treatment strategies can even make things worse.

5.4.2 Guideline to prevent early stage failures

Basically, one has to distinguish between initial commissioning and retrofit and modification. Any initial commissioning is accompanied by a preceding inspection phase over a period of up to several months. During this time, it can be assured that occurring early stage failures are revealed. There is also the possibility to run in the spare parts and reject the faulty ones.

After commissioning the system must not be used for test purposes anymore. That means that the reliability of replenished spare parts must be ensured by other means. In order to guarantee this level of reliability the

permanent employment of all components before putting an installation into service has to be planned and monitored by the RAM Management. Furthermore, a failure reporting system must be set up to verify the level of quality of the items employed. These records should be used for further considerations on pre-treatment strategies.

When adding new components to a specific system, this observation period is not given. In this and in all other cases where an inspection phase is not possible, further considerations are needed and have to be defined during the requirement process of the system. The decision finding should be based on a cost-benefit analysis. These cost-benefit considerations should take into account the resulting operational consequences in case of failure.

5.5 Spare parts policy

5.5.1 Spare parts stocking

5.5.1.1 General

Basically, the number of spare parts proposed for specific units has to be seen merely as a proposal from GTS. The decision on the real number of spare parts belongs to the business of the railway duty holder.

5.5.1.2 Number of spare parts

The determination must be done in accordance with the standard “spare parts calculation” [7]. The number of spare parts of a specific unit is based on the corresponding failure rate and an additional NFF share (No Fault Found), the number of units employed, the confidence level and the so-called “Turn Around Time”.

The choice of the degree of the confidence level determines the residual risk, which one is willing to accept that there is no spare part left in case of fault. A confidence level of 98% as well as an additional NFF-share of 20% should be taken. Nevertheless, these values can be varied depending on the task given.

5.5.1.3 Tools

The number of spare parts can be calculated with the spare parts calculation tool or a similar tool.

5.5.2 Periodical change of spare parts

The necessity for a periodical inspection or change of spare parts shall be avoided. Deviations from this rule have to be explicitly justified.

The preconditions allowing normal operation with spare parts must already be fulfilled.

These are

- Full functioning
- Acceptable percentage of early stage failures.

5.6 Field statistics

5.6.1 Failure reporting

A failure reporting and corrective action system (FRACAS) has to be set up in order to be able to evaluate all hardware failures as well as software errors during operation. In addition, all corrective measures taken have to be recorded.

5.6.1.1 Repair logistics

Basically, the faulty repairable units should be sent to a central institution – the repair centre. This repair centre electronically registers the units and sends them to the corresponding manufacturer for repair. After successful repair a repair report has to be created and archived.

5.6.1.2 Repair statistics

The resulting failure data are used by the RAM Management to calculate the empirical failure rates of single printed board assemblies and devices, whereby the total number of field units serves as a basis.

The statistics also should enable the evaluation of single installations and lines in order to be able to detect local anomalies. To quickly react to certain component anomalies the repair statistics also should allow a tracing up to the component level (e.g. increased failure frequencies of electrolytic capacitors after a certain operation time).

Basically, in the repair statistics 3 failure categories have to be distinguished:

- Randomly behaving failures
- Systematic failures (from design faults, manufacturing faults, etc.)
- Failures due to external influences (Lightning etc.)

A further category covers cases, where units are mistakenly supposed to be defect and therefore sent to the repair centre. Generally, these units are called “No Fault Found Units”. For this special group a procedure is defined to prevent that an NFF-unit will loop several times through the whole repair process (see [S11]).

Failures during the run-in phase and failures during normal operation have to be treated separately. Doing so, it is possible to verify the quality of the functional test as well as to identify early stage failures. In addition to that the time to failure of each failed unit should be recorded.

A proper procedure for evaluating the number of failures of non-repairable units that are not processed by the repair centre has to be defined. This procedure could be based on supplier data.

In addition to the repair statistics the so-called „Turn Around Time“ of each unit must be recorded. This also holds for non-repairable units.

5.6.2 Evaluation of field data

In order to be able to compare the empirical data with the predicted failure rates one has to take into account the fault class already defined in chapter 5.3.2.2.

For calculating the empirical failure rate the following parameters are necessary:

- Observation time
- Number of units failed within the observation time
- Number of units at the beginning of the observation period.

As a rule, the use of point estimators should be avoided. Instead of stating a single figure, a confidence interval should be used. This interval represents the area, in which the true value falls at a certain level of confidence. The default confidence level should be 98%.

The determination of the confidence intervals must be based on exponentially distributed lifetime [S10].

5.6.3 Software errors

Faults of a systematic nature such as software design faults are minimized by following the rules of CENELEC during the development process. From the railway duty holders’ point of view, it is becoming more and more important that the suppliers do not only predict system failures which can be attributed to hardware faults, but also make statements concerning software reliability of the product. That means that the quantitative reliability requirements stated also hold for system failures caused by software errors.

Basically, it should be avoided putting random hardware failures and software errors in the same failure category and special care should be taken in forecasting software reliability figures.

However, in order to be prepared software errors and the associated operational consequences of any railway signalling facilities in the field have to be collected and adequately evaluated.

5.7 RAM-based LCC

5.7.1 General

Generally, the life cycle cost consists of a large cost-chain including purchasing, operational, inspection, corrective and preventive maintenance costs over the planned product lifetime (see [9]).

The RAM-based LCC are

- Purchasing costs of spare parts
- Repair/Replacement costs of LRUs
- Inspection and maintenance costs
- System down time costs.

5.7.2 Repair, Inspection and Maintenance Costs

The failures to be expected during the planned product lifetime are based on constant failure rates. Inspection and maintenance activities subsume material costs as well as personnel costs for the operators. These costs vary on the number of maintenance actions, the average time needed per action and the hourly wage rate of the maintenance staff. There is also need for information on the hourly wage rates of the maintenance staff to predict this portion of costs.

Typically, the calculations take into account a price index and a discount factor with payment in arrear, both annually.

5.7.3 Spare Parts Costs

The calculation of the required number of spare parts is based on the so-called turn around time (TAT). The turn around time is the time span starting with the failure of a unit and the replacement by a spare part until the repaired part is available on stock again.

The number of spare parts of individual components has to be calculated at a confidence level of 98% including an additional amount of 20% for components, which have been mistakenly supposed to be faulty.

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